

making a mockery of some poorly powered desktops. This model comes built with an ATI M6-P Integrated AGP 4X graphics card which has a fully DirectX 8 compliant graphics engine, coupled with 32 MB of ultra fast DDR memory.

Direct3D functionality was pretty decent too, and some notebooks like the Compaq Presario 1725 even went on to complete the 3DMARK 2000 benchmark.

Just goes to show that if you are in the mood for that odd game or two while traveling at 35,000 feet altitude, you won't be disappointed.

Battery power

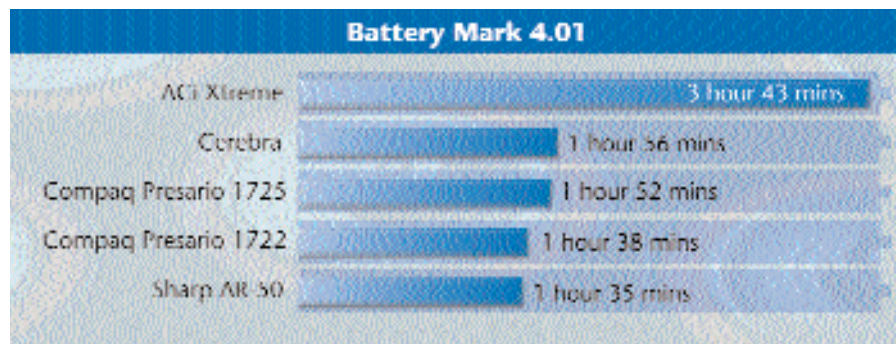
The ability to put your laptop on in a remote location with no electrical source is true computing freedom. NiCad (Nickel Cadmium) and NiMH (Nickel Metal-Hydrate) were the more popular types of batteries used in laptops. These have now been replaced by the ever-popular Li-Ion (Lithium-Ion) batteries, which provide a longer life and cannot be overcharged. This makes it ideal for laptops. Also, due to higher density, these batteries are smaller and lighter than NiMH batteries.

All the laptops came fitted with lithium ion batteries. To drain the notebook's battery, we ran BatteryMark 4.01, which uses the tests of Business Winstone 2001. This benchmark is scripted in such a way that it runs applications which drain a notebook's battery—it mimics real-time usage.

We warmed up the notebook by running it for an hour; the battery was then left to charge overnight. All power management software was disabled and the notebook was then disconnected from the mains. Expect an extra 10-20 per cent battery performance if you have power management enabled.

Almost all the notebooks had a battery life of about one-and-a-half hours under this test. Except for one, the ACi Xtreme truly took battery power to an 'extreme'. This notebook comes specially fitted with a 5-hour rated battery and it ran admirably in our tests for an amazing 3 hours 43 minutes. Using power management, this notebook can easily cross 4 hours of nonstop usage. The ACi Xtreme comes with another 5-hour battery as an extra—this gives you nonstop usage of over 8 hours!

Another notable mention was the Cerebra notebook, which came close to



crossing the 2-hour barrier in non-stop usage by scoring 1 hour 56 minutes. The Compaq Presario 1725 scored a respectable 1 hour 52 minutes.

I/O functionality, features and expandability

Laptops also come with additional docking connections which allow users to connect an extra monitor, keyboard, mouse, printer, etc to the laptop and use it as a regular desktop PC. Most computers have the infrared port so that other infrared devices like PDA can be used with the laptop.

The I/O functionality of the laptops was tested against a minimum requirement of certain ports and features.

Most of the laptops fulfilled our minimum criterion and were well equipped. Notably, both the Compaq Presario notebooks featured an IEEE 1394 FireWire port.

This port also came standard with the Sharp AR-50, ACi Xtreme and the ACi Profile 3270.

The ACi Xtreme, Zenith One up, Sharp AR-50 and the Compaq Armada E-500 also came with two PCMCIA Type II slots. The rest came with only one.

In terms of raw features the Compaq Presario 1725 towered over others, featuring JBL Pro Audio with Bass reflex and Dolby Digital certified sound that was hard to beat. The musical output of this notebook was truly inspiring, a far cry from the tiny speakers that came with the remaining models. Nevertheless, the ACi Xtreme wasn't far behind: it featured one Sony standard Memory Stick along with one Panasonic SD Memory card, support for optional 802.11b Wireless LAN, and a 3D enhanced sound system using a Sound-Blaster PRO interface. The ACi Xtreme that we received had a DVD-ROM drive attached. The Compaq Presario 1725, on the other hand, came fitted with a hot swappable 4x4x24x6xCD-RW-DVD combo drive. The Sharp AR-50 too had an 8x/4x/24x CD-Writer.

Quality and display size

The thin flat LCD (Liquid Crystal Diode) display on a laptop or notebook is the most delicate of all components, and also a major chunk of the cost component of a notebook. While the performance wasn't comparable to CRTs earlier on, the latest developments in size and transistor response times have made displays pretty comparable. Bad transistors and pixels are a part and parcel of the entire laptop experience. Though this problem has been fixed to a certain degree with more expensive notebooks, you still need to be

